Financing Frictions: Empirics (Basics)

Adrien Matray
(Federal Reserve Bank of Atlanta, CEPR)

PhD Corporate Finance

Roadmap: each method fixes the previous one's identification flaw

- Cross-country correlation (King & Levine)
  - ✗ Omitted variables and reverse causality
- Rajan & Zingales: Within-country, cross-industry fixed effects
  - ✗ Country FE absorbs country-level finance: need within-country variation
  - ✗ $FinDev_c$ and EF_i are equilibrium objects, correlated with unobservables
- Bertrand, Schoar & Thesmar: banking-deregulation natural experiment
  - ✓ Design-based: identification rests on parallel trends

Outline

Basics of DiD, finance, and growth

Challenges in identifying financing frictions

Outline

Every measure of finance correlates with growth

"Finance and Growth: Schumpeter Might Be Right", King & Levine, QJE 1993

Figure: Correlation between financial development & growth

	Very fast	Fast	Slow	Very slow	Correlation with capital growth	(*P*-value)
LLY	0.65	0.38	0.24	0.21	0.69	(0.001)
BANK	0.88	0.75	0.64	0.60	0.57	(0.001)
PRIVATE	0.73	0.62	0.54	0.50	0.50	(0.001)
PRIVY	0.43	0.23	0.16	0.14	0.65	(0.001)
GK	0.014	0.001	−0.007	−0.021		

Very fast: $GK > 0.0072$, Fast: $GK > -0.0022$ and < 0.0072, Slow: $GK > -0.0126$ and < -0.0022, Very slow: $GK < -0.0126$.

LLY = Ratio of liquid liabilities to GDP, *BANK* = Deposit money bank domestic credit divided by deposit money bank plus central bank domestic credit, *PRIVATE* = Ratio of claims on the nonfinancial private sector to total domestic credit, *PRIVY* = Ratio of claims on the nonfinancial private sector to GDP, *GK* = Average growth rate of the real per capita capital stock, 1960–1989.

Observations: Approximately twenty in each of the four categories.

Fin depth: LLY = financial sector liabilities / GDP

Intermediated finance: BANK = bank assets /(bank+central bank assets) ; PRIVATE = bank loans to non financial sectors / total credit ;

Asset allocation: PRIVY = loans to non financial industries/GDP

GDP growth and financial development are correlated

$$\Delta \log GDP_c = \alpha + \underbrace{\beta}_{>0} FinDev_c + \varepsilon_c$$

$\rightarrow$ So what?

GDP growth and financial development are correlated

$$\Delta \log GDP_c = \alpha + \underbrace{\beta}_{>0} FinDev_c + \varepsilon_c$$

$\rightarrow$ So what?

1. Omitted variable (OVB)

GDP growth and financial development are correlated

$$\Delta \log GDP_c = \alpha + \underbrace{\beta}_{>0} FinDev_c + \varepsilon_c$$

$\rightarrow$ So what?

1. Omitted variable (OVB)

2. Reverse causality

   - True model: $FinDev_c = a_2 + b_2 \Delta \log GDP_c + v_c$

   $\rightarrow$ Does it help to lag $FinDev_c$?

Within-country variation controls for unobserved country traits

- Countries vary on too many dimensions + hard to measure

  $\rightarrow$ Country-specific unobserved factors affecting both financial development and growth

- Solution: within country $\neq$ across countries variations in financial development

Demeaning within country removes the country fixed effect

Suppose the true model is

$$\Delta \log VA_{i,c} = \alpha_c + \beta FinDev_{i,c} + \varepsilon_{i,c}$$

where α_c is an unobserved country characteristic

Demeaning within country removes the country fixed effect

Suppose the true model is

$$\Delta \log VA_{i,c} = \alpha_c + \beta FinDev_{i,c} + \varepsilon_{i,c}$$

where α_c is an unobserved country characteristic

Averaging the previous equation across industries within each country:

$$\overline{\Delta \log VA_{i,c}} = \alpha_c + \beta \overline{FinDev_{i,c}} + \overline{\varepsilon_{i,c}}$$

Demeaning within country removes the country fixed effect

Suppose the true model is

$$\Delta \log VA_{i,c} = \alpha_c + \beta FinDev_{i,c} + \varepsilon_{i,c}$$

where α_c is an unobserved country characteristic

Averaging the previous equation across industries within each country:

$$\overline{\Delta \log VA_{i,c}} = \alpha_c + \beta \overline{FinDev_{i,c}} + \overline{\varepsilon_{i,c}}$$

Hence

$$\Delta \log VA_{i,c} - \overline{\Delta \log VA_{i,c}} = \beta(FinDev_{i,c} - \overline{FinDev_{i,c}}) + \varepsilon_{i,c} - \overline{\varepsilon_{i,c}}$$

Demeaning within country removes the country fixed effect

Suppose the true model is

$$\Delta \log VA_{i,c} = \alpha_c + \beta FinDev_{i,c} + \varepsilon_{i,c}$$

where α_c is an unobserved country characteristic

Averaging the previous equation across industries within each country:

$$\overline{\Delta \log VA_{i,c}} = \alpha_c + \beta \overline{FinDev_{i,c}} + \overline{\varepsilon_{i,c}}$$

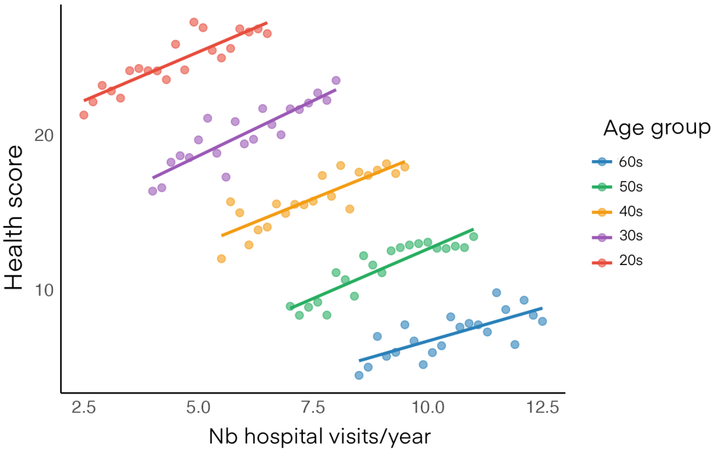
Hence

$$\Delta \log VA_{i,c} - \overline{\Delta \log VA_{i,c}} = \beta(FinDev_{i,c} - \overline{FinDev_{i,c}}) + \varepsilon_{i,c} - \overline{\varepsilon_{i,c}}$$

Demeaning removes the country-level unobserved heterogeneity and allows to identify β

This is called a fixed effects estimator

Fixed effects recover the within-group relationship



Within-country, cross-industry variation sharpens identification

$$\Delta \log VA_c = \alpha_c + FinDev_c + \varepsilon_c$$

- Country FE $\rightarrow$ removes unobserved heterogeneity across countries $\rightarrow$ Pb?

Within-country, cross-industry variation sharpens identification

$$\Delta \log VA_c = \alpha_c + FinDev_c + \varepsilon_c$$

- Country FE $\rightarrow$ removes unobserved heterogeneity across countries $\rightarrow$ Pb?

- Solution: create *within*-country variation. Industry i, country c:

$$\Delta \log VA_{c,i} = \alpha_c + FinDev_c + \varepsilon_{c,i}$$

$\rightarrow$ Pb?

Rajan-Zingales: finance matters more for finance-dependent industries

- Define $FinDev_c$ as financial development in country c and estimate:

$$\Delta \log VA_{i,c} = \alpha_c + \alpha_i + \beta FinDev_c + \varepsilon_{i,c}$$

$\rightarrow$ Pb? Country FE α_c absorbs $FinDev_c$: β not identified

Rajan-Zingales: finance matters more for finance-dependent industries

- Define $FinDev_c$ as financial development in country c and estimate:

$$\Delta \log VA_{i,c} = \alpha_c + \alpha_i + \beta FinDev_c + \varepsilon_{i,c}$$

$\rightarrow$ Pb? Country FE α_c absorbs $FinDev_c$: β not identified

- Solution: create within-country variation. Hypothesis: financial development should matter more in industries that need external funds to run their business
  - Define EF_i external finance dependence of industry i

$$\Delta \log VA_{i,c} = \alpha_c + \alpha_i + \beta FinDev_c \times EF_i + \varepsilon_{i,c}$$

- How to measure external finance dependence EF_i? Use US firms as the benchmark:

$$EF_i = \left(\frac{\text{investment} - \text{cash flows from operations}}{\text{investment}} \right)_{i,US}$$

using accouting data from COMPUSTAT

Rajan-Zingales: finance matters more for finance-dependent industries

Solution: create within-country variation

$\rightarrow$ EF_i external finance dependence of industry i

$$\Delta \log VA_{i,c} = \alpha_c + \alpha_i + \beta FinDev_c \times EF_i + \varepsilon_{i,c}$$

- How to measure external finance dependence EF_i? Use US firms as the benchmark:

$$EF_i = \left(\frac{\text{investment} - \text{cash flows from operations}}{\text{investment}} \right)_{i,US}$$

  using accouting data from COMPUSTAT

Finance speeds growth most where firms depend on external funds

Variable	Financial development measured as					
	Total capitalization	Bank debt	Accounting standards	Accounting standards in 1983	Accounting standards and capitalization	Instrumental variables
Industry's share of total value added in manufacturing in 1980	−0.912 (0.246)	−0.899 (0.245)	−0.643 (0.204)	−0.587 (0.223)	−0.443 (0.135)	−0.648 (0.203)
Interaction (external dependence × total capitalization)	0.069 (0.023)	—	—	—	0.012 (0.014)	—
Interaction (external dependence × domestic credit to private sector)	—	0.118 (0.037)	—	—	—	—
Interaction (external dependence × accounting standards)	—	—	0.155 (0.034)	—	0.133 (0.034)	0.165 (0.044)
Interaction (external dependence × accounting standards 1983)	—	—	—	0.099 (0.036)	—	—
R^2	0.290	0.290	0.346	0.239	0.419	0.346
Number of observations	1217	1217	1067	855	1042	1067
Differential in real growth rate	1.3	1.1	0.9	0.4	1.3	1.0

(Machinery-Beverages) in Italy vs. (Machinery-Beverages) in Philippines

Finance speeds growth most where firms depend on external funds

	Financial development measured as					
Variable	Total capitalization	Bank debt	Accounting standards	Accounting standards in 1983	Accounting standards and capitalization	Instrumental variables
Industry's share of total value added in manufacturing in 1980	−0.912 (0.246)	−0.899 (0.245)	−0.643 (0.204)	−0.587 (0.223)	−0.443 (0.135)	−0.648 (0.203)
Interaction (external dependence × total capitalization)	$\boxed{0.069\ (0.023)}$	—	—	—	0.012 (0.014)	—
Interaction (external dependence × domestic credit to private sector)	—	0.118 (0.037)	—	—	—	—
Interaction (external dependence × accounting standards)	—	—	0.155 (0.034)	—	0.133 (0.034)	0.165 (0.044)
Interaction (external dependence × accounting standards 1983)	—	—	—	0.099 (0.036)	—	—
R^2	0.290	0.290	0.346	0.239	0.419	0.346
Number of observations	1217	1217	1067	855	1042	1067
Differential in real growth rate	$\boxed{1.3}$	1.1	0.9	0.4	1.3	1.0

(Machinery-Beverages) in Italy vs. (Machinery-Beverages) in Philippines

Remaining concerns?

Bertrand-Schoar-Thesmar: a banking-deregulation natural experiment

- Endogeneity concerns in Rajan & Zingales 1998 come from $FinDev_c$ and EF_i being equilibrium objects, correlated with many unobservables
- "Natural experiment": move to a design-based approach

Bertrand-Schoar-Thesmar: a banking-deregulation natural experiment

- Endogeneity concerns in Rajan & Zingales 1998 come from $FinDev_c$ and EF_i being equilibrium objects, correlated with many unobservables
- "Natural experiment": move to a design-based approach

- Country-level banking deregulation in 1986 + universe of firms' tax files
  → Firm-level data now: i = firm, j = industry, c = city, t = year

$$\log K_{i,j,c,t} = \gamma_j + \alpha_i + \beta Deregulation_t + \delta_t + \varepsilon_{i,j,c,t}$$

→ Pbs?

Dosage: interact the shock with industry finance dependence

Firm i in industry j in city c at time t

$$\log K_{i,j,c,t} = \alpha_i + \beta \mathit{Deregulation}_t \times \mathit{FinDev}_j + \delta_t + \varepsilon_{i,j,c,t}$$

Dosage: interact the shock with industry finance dependence

Firm i in industry j in city c at time t

$$\log K_{i,j,c,t} = \alpha_i + \beta \mathit{Deregulation}_t \times \mathit{FinDev}_j + \delta_t + \varepsilon_{i,j,c,t}$$

Some questions:

Dosage: interact the shock with industry finance dependence

Firm i in industry j in city c at time t

$$\log K_{i,j,c,t} = \alpha_i + \beta Deregulation_t \times FinDev_j + \delta_t + \varepsilon_{i,j,c,t}$$

Some questions:

1. Replace α_i by γ_j $\rightarrow$ Can we have $\beta^{\alpha_i} = \beta^{\gamma_j}$?

Dosage: interact the shock with industry finance dependence

Firm i in industry j in city c at time t

$$\log K_{i,j,c,t} = \alpha_i + \beta Deregulation_t \times FinDev_j + \delta_t + \varepsilon_{i,j,c,t}$$

Some questions:

1. Replace α_i by γ_j → Can we have $\beta^{\alpha_i} = \beta^{\gamma_j}$?

2. With α_i, does it matter to remove firm entering post deregulation? Firm exiting pre-deregulation?

Dosage: interact the shock with industry finance dependence

Firm *i* in industry *j* in city *c* at time *t*

$$\log K_{i,j,c,t} = \alpha_i + \beta Deregulation_t \times FinDev_j + \delta_t + \varepsilon_{i,j,c,t}$$

Some questions:

1. Replace α_i by $\gamma_j \rightarrow$ Can we have $\beta^{\alpha_i} = \beta^{\gamma_j}$?

2. With α_i, does it matter to remove firm entering post deregulation? Firm exiting pre-deregulation?

3. We worry about potential unobserved shocks at the industry level, can we include $\gamma_j \times \delta_t$?

Dosage: interact the shock with industry finance dependence

Firm i in industry j in city c at time t

$$\log K_{i,j,c,t} = \alpha_i + \beta \mathit{Deregulation}_t \times \mathit{FinDev}_j + \delta_t + \varepsilon_{i,j,c,t}$$

Some questions:

1. Replace α_i by $\gamma_j \rightarrow$ Can we have $\beta^{\alpha_i} = \beta^{\gamma_j}$?

2. With α_i, does it matter to remove firm entering post deregulation? Firm exiting pre-deregulation?

3. We worry about potential unobserved shocks at the industry level, can we include $\gamma_j \times \delta_t$?

4. How is β identify if we include $\mu_c \times \delta_t$?

Outline

Basics of DiD, finance, and growth

- What does the data say?
- Method: Identifying assumption
- Method: Interpreting coefficients

Challenges in identifying financing frictions

- Method: Triple differences
- How to identify a financially constrained firm?

Parallel trends is the identifying assumption for DD

- Parallel trends: absent treatment, treated and control would have followed the same change in expected outcome

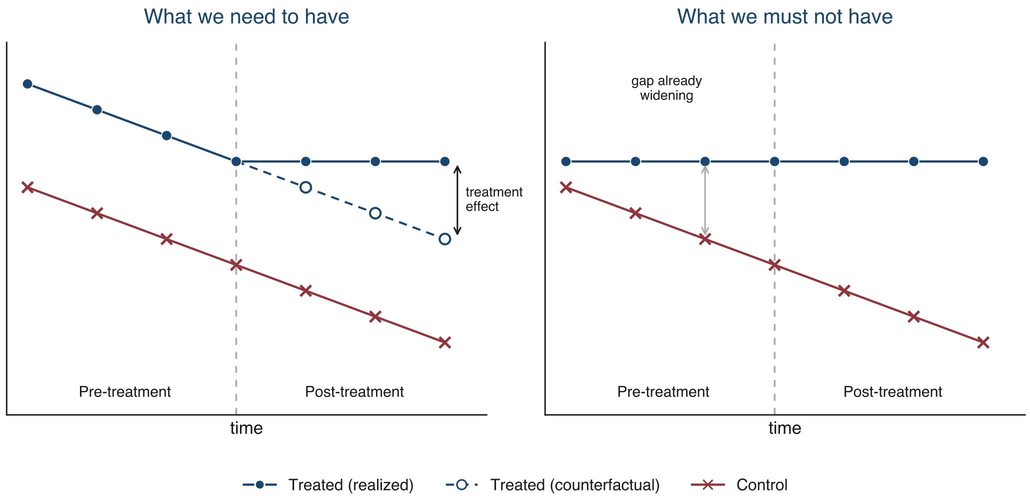
  $\rightarrow$ $E\left[Y_t^0 - Y_{t-1}^0 \mid \text{Treated}\right] = E\left[Y_t^0 - Y_{t-1}^0 \mid \text{Control}\right]$

  About the unobserved counterfactual Y^0: untestable directly

- What it does not imply
  - Not imply: no trend in the outcome before the treatment (the trends must just be similar across groups
  - Not imply: similar level of the outcome
    - Though similar support makes it more plausible

    $\rightarrow$ Show us the balance covariate test!!!

What parallel trends needs, and what violates it



Two challenges: covariate imbalance and treated-specific shocks

$$\log(\textit{Capital}_{i,t}) = \beta\, \textit{Banking Reform}_i \times \textit{Post}_{t>2010} + \alpha_i + \delta_t$$

Two challenges: covariate imbalance and treated-specific shocks

Challenge 1: Imbalance on covariates $\Rightarrow$ aggregate shocks load differently on treated

$$\log(Capital_{i,t}) = \beta \, Banking\ Reform_i \times Post_{t>2010} + \alpha_i + \delta_t$$
$$+ \gamma \, X_i \times Post_{t>2010}$$
$$\longrightarrow \text{Problem if } \; Cov(X_i, \; Banking\ Reform_i) \neq 0$$

Two challenges: covariate imbalance and treated-specific shocks

Challenge 1: Imbalance on covariates $\Rightarrow$ aggregate shocks load differently on treated

$$\log(Capital_{i,t}) = \beta\, Banking\ Reform_i \times Post_{t>2010} + \alpha_i + \delta_t$$

$$+ \gamma\, X_i \times Post_{t>2010}$$

$\longrightarrow$ Problem if $Cov(X_i,\ Banking\ Reform_i) \neq 0$

- Solutions:
  - Show balance covariate test
  - Use additional controls to tighten identification
    - Potentially disaggregate the data for higher-dimension FE $\Rightarrow$ how to disaggregate?
    - Control for source of imbalance $\Rightarrow$ risk of mean reversion (more on this later)

Two challenges: covariate imbalance and treated-specific shocks

Challenge 2: Treated units have specific additional shocks

$$\log(\mathit{Capital}_{i,t}) = \beta\, \mathit{Banking\ Reform}_i \times \mathit{Post}_{t>2010} + \alpha_i + \delta_t$$

$$+ \theta\, \mathit{Trade\ Shock}_i \times \mathit{Post}_{t>2010}$$

$$\longrightarrow \hat{\beta} = \beta + \theta \;\; \text{not separable}$$

Two challenges: covariate imbalance and treated-specific shocks

Challenge 2: Treated units have specific additional shocks

$$\log(\mathit{Capital}_{i,t}) = \beta\, \mathit{Banking\ Reform}_i \times \mathit{Post}_{t>2010} + \alpha_i + \delta_t$$
$$+\, \theta\, \mathit{Trade\ Shock}_i \times \mathit{Post}_{t>2010}$$
$$\longrightarrow \hat{\beta} = \beta + \theta \;\; \text{not separable}$$

- Solutions: can't control (collinear) $\Rightarrow$ need new variation
  - Triple difference: a group with the trade shock but not the reform
  - Placebo: outcome moved by trade but not credit

Outline

Basics of DiD, finance, and growth

- What does the data say?
- Method: Identifying assumption
- Method: Interpreting coefficients

Challenges in identifying financing frictions

- Method: Triple differences
- How to identify a financially constrained firm?

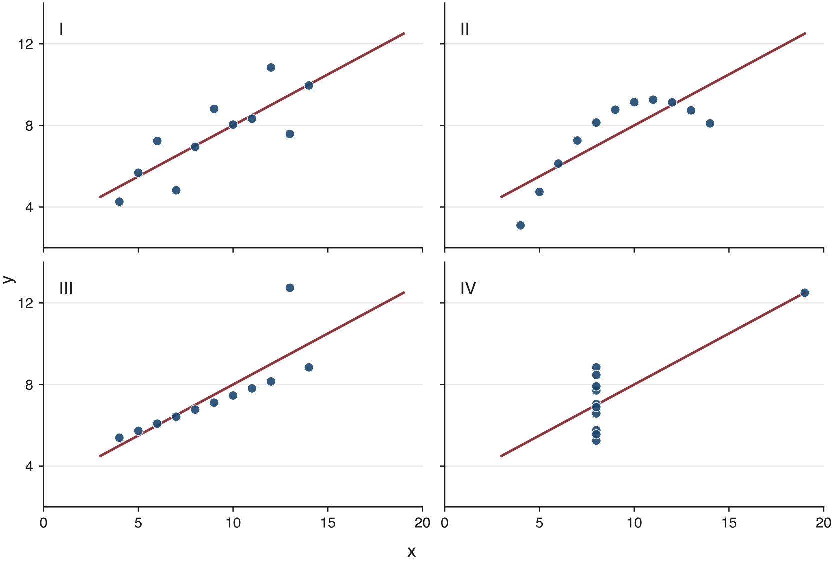
DD is an IV: show the first stage and cluster at the shock

- DD $\approx$ IV regression
  - Experiment = significant change in economic environment
  - Try to show the "first stage"
- Show the graph! (and be suspicious of papers which do not)
- Clusters (Bertrand et al. 2004): panel so cluster ideally at the level of shock (states, industry, etc.)

Reduced-form DD is hard to compare and rescale

- Harder to compare studies
  - e.g. "foreign capital liberalization" $\rightarrow$ debt (bond? bank?)? equity (ST, LT)?
- Rescaling is not always possible
  - Upward bias if shock changes multiple variables at the same time
    - equivalent to exclusion restriction violated
  - Shock might be the true effect
    - e.g. Fonseca & Matray (2023)

Show the graph!



Event-study shapes: jump, perpetual rise, or plateau

Three types:

1. The jump post shock

2. The progressive and perpetual increase

3. The progressive then plateau

Log versus level changes what the coefficient means

$$Log(Capital_{i,t}) = \beta \mathbb{1}[Banking\ Reform_i = 1] \times Post_{t>2010} + \alpha_i + \delta_t$$

Log versus level changes what the coefficient means

$$Log(Capital_{i,t}) = \beta \mathbb{1}[Banking\ Reform_i = 1] \times Post_{t>2010} + \alpha_i + \delta_t$$

- Estimate: Log(Capital) with $\beta = 0.1 \rightarrow$ Interpretation?

Log versus level changes what the coefficient means

$$Log(Capital_{i,t}) = \beta \mathbb{1}[Banking\ Reform_i = 1] \times Post_{t>2010} + \alpha_i + \delta_t$$

- Estimate: Log(Capital) with $\beta = 0.1 \rightarrow$ Interpretation?
  - NB: Log creates challenges when unpacking $Y_{c,t}$: Zeros and Jensen inequality
  - $K_{i,t} = PPE_{i,t} + Intan_{i,t}$,   but: $\beta^{log(K)} \neq \beta^{log(PPE)} + \beta^{log(Intan)}$

Log versus level changes what the coefficient means

$$Log(Capital_{i,t}) = \beta \mathbb{1}[Banking\ Reform_i = 1] \times Post_{t>2010} + \alpha_i + \delta_t$$

- Estimate: Log(Capital) with $\beta = 0.1 \rightarrow$ Interpretation?
  - NB: Log creates challenges when unpacking $Y_{c,t}$: Zeros and Jensen inequality
  - $K_{i,t} = PPE_{i,t} + Intan_{i,t}$, but: $\beta^{log(K)} \neq \beta^{log(PPE)} + \beta^{log(Intan)}$

- Estimate: Capital with $\beta = 100 \rightarrow$ Interpretation?

Log versus level changes what the coefficient means

$$Log(Capital_{i,t}) = \beta \mathbb{1}[Banking\ Reform_i = 1] \times Post_{t>2010} + \alpha_i + \delta_t$$

- Estimate: Log(Capital) with $\beta = 0.1 \rightarrow$ Interpretation?
  - NB: Log creates challenges when unpacking $Y_{c,t}$: Zeros and Jensen inequality
  - $K_{i,t} = PPE_{i,t} + Intan_{i,t}$, but: $\beta^{log(K)} \neq \beta^{log(PPE)} + \beta^{log(Intan)}$
- Estimate: Capital with $\beta = 100 \rightarrow$ Interpretation?
  - NB: Potential bias when estimating $Y_{c,t}$ in level: Granularity $\Rightarrow$ LLN does not hold
    - $\rightarrow$ See lecture "Empirics: advanced"

Outline

Basics of DiD, finance, and growth

Challenges in identifying financing frictions

Outline

Basics of DiD, finance, and growth

- What does the data say?
- Method: Identifying assumption
- Method: Interpreting coefficients

Challenges in identifying financing frictions

- Method: Triple differences
- How to identify a financially constrained firm?

Interaction term

$$Log(Capital_{i,t}) = \beta DiD_{i,t} \mathbb{1}[Banking\ Reform_i = 1] \times Post_{t>2010} + \alpha_i + \delta_t$$

Might want to study additional cross-sectional variation

$$Log(Capital_{i,t}) = \beta_1 DiD_{i,t} + \beta_2 DiD_{i,t} \times \mathbb{1}[Large_{i,t0}] + \alpha_i + \delta_t$$

Two elements to consider:

1. Interpreting β_1 and β_2

2. Classic mistake with triple diff

Interaction term: interpretation

$$Log(Capital_{i,t}) = \beta_1 DiD_{i,t} + \beta_2 DiD_{i,t} \times \mathbb{1}[Large_{i,t0}] + \alpha_i + \delta_t$$

- β_1 = effect of the shock when interaction ($[Large_{i,t0}]$) = 0

  $\rightarrow$ What if interaction is continuous?

Interaction term: interpretation

$$Log(Capital_{i,t}) = \beta_1 DiD_{i,t} + \beta_2 DiD_{i,t} \times \mathbb{1}[Large_{i,t0}] + \alpha_i + \delta_t$$

- β_1 = effect of the shock when interaction ($[Large_{i,t0}]$) = 0

  $\rightarrow$ What if interaction is continuous?

- β_2 = marginal effect

  $\rightarrow$ Total effect of shock on large firms?

Interaction term: standard mis-specified regression

- **All** RHS should be interacted with the interaction term

Interaction term: standard mis-specified regression

- All RHS should be interacted with the interaction term
- When in doubt
  - Run the conditional regression
  - Make sure you retrieve the same coeff

Mispecification can completely flip the conclusion

Interaction term: example of mis-specification

Estimate:

$$\frac{I_{i,t}}{PPE_{i,t-1}} = log(Age_{i,t}) + \alpha_i + \delta_t$$

Separately for:

- $\mathbb{1}[Big_{i,t0}] = 0$
- $\mathbb{1}[Big_{i,t0}] = 1$

```
. reghdfe capex1 lage if big==0, a(gvkey year) vce(cluster gvkey)
(dropped 115 singleton observations)
(converged in 8 iterations)

HDFE Linear regression                            Number of obs   =     12,474
Absorbing 2 HDFE groups                           F(   1,    1632) =     222.93
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.3155
                                                  Adj R-squared   =     0.2113
                                                  Within R-sq.    =     0.0430
Number of clusters (gvkey)   =      1,633         Root MSE        =     0.5725

                           (Std. Err. adjusted for 1,633 clusters in gvkey)
------------------------------------------------------------------------------
             |               Robust
      capex1 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        lage |  -.5486915   .0367486   -14.93   0.000     -.620771   -.476612
------------------------------------------------------------------------------
. reghdfe capex1 lage if big==1, a(gvkey year) vce(cluster gvkey)
(dropped 90 singleton observations)
(converged in 8 iterations)

HDFE Linear regression                            Number of obs   =     12,531
Absorbing 2 HDFE groups                           F(   1,    1345) =      82.36
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.4176
                                                  Adj R-squared   =     0.3467
                                                  Within R-sq.    =     0.0286
Number of clusters (gvkey)   =      1,346         Root MSE        =     0.2715

                           (Std. Err. adjusted for 1,346 clusters in gvkey)
------------------------------------------------------------------------------
             |               Robust
      capex1 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        lage |  -.2493993    .027482    -9.07   0.000    -.3033116    -.195487
------------------------------------------------------------------------------
```

Interaction term: example of mis-specification

Estimate:

$$\frac{I_{i,t}}{PPE_{i,t-1}} = log(Age_{i,t}) + \alpha_i + \delta_t$$

Separately for:

- $\mathbb{1}[Big_{i,t0}] = 0$
- $\mathbb{1}[Big_{i,t0}] = 1$

Investment to age sensitivity should be roughly half muted for large firms

- $\beta^{big} = -0.2493993$
- $\beta^{small} = -0.5486915$

$\Rightarrow$ Marginal effect = $=$ 0.2992922

```
. reghdfe capex1 lage if big==0, a(gvkey year) vce(cluster gvkey)
(dropped 115 singleton observations)
(converged in 8 iterations)

HDFE Linear regression                            Number of obs   =     12,474
Absorbing 2 HDFE groups                           F(   1,   1632) =     222.93
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.3155
                                                  Adj R-squared   =     0.2113
                                                  Within R-sq.    =     0.0430
Number of clusters (gvkey)   =      1,633         Root MSE        =     0.5725

                              (Std. Err. adjusted for 1,633 clusters in gvkey)
------------------------------------------------------------------------------
             |               Robust
      capex1 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        lage |  -.5486915   .0367486   -14.93   0.000     -.620771    -.476612
------------------------------------------------------------------------------
. reghdfe capex1 lage if big==1, a(gvkey year) vce(cluster gvkey)
(dropped 90 singleton observations)
(converged in 8 iterations)

HDFE Linear regression                            Number of obs   =     12,531
Absorbing 2 HDFE groups                           F(   1,   1345) =      82.36
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.4176
                                                  Adj R-squared   =     0.3467
                                                  Within R-sq.    =     0.0286
Number of clusters (gvkey)   =      1,346         Root MSE        =     0.2715

                              (Std. Err. adjusted for 1,346 clusters in gvkey)
------------------------------------------------------------------------------
             |               Robust
      capex1 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        lage |  -.2493993    .027482    -9.07   0.000    -.3033116    -.195487
------------------------------------------------------------------------------
```

Interaction term: example of mis-specification

Run naive triple-diff

$$\frac{I_{i,t}}{PPE_{i,t-1}} = log(Age_{i,t}) \times [Big_{i,t0}] + log(Age_{i,t}) + \alpha_i + \delta_t$$

Remember that we should have:

- $\beta^{small} = -0.5486915$

$\Rightarrow$ Marginal effect = $= 0.2992922$

Interaction term: example of mis-specification

Run naive triple-diff

$$\frac{I_{i,t}}{PPE_{i,t-1}} = log(Age_{i,t}) \times [Big_{i,t0}] + log(Age_{i,t}) + \alpha_i + \delta_t$$

Remember that we should have:

- $\beta^{small} = -0.5486915$
  - -0.4259

$\Rightarrow$ Marginal effect = $= 0.2992922$

  - 0.0546

```
. reghdfe capex1 big lage lage_big , a(gvkey year) vce(cluster gvkey)
(dropped 38 singleton observations)
(converged in 8 iterations)

HDFE Linear regression                            Number of obs   =     25,172
Absorbing 2 HDFE groups                           F(   3,   2520) =     117.91
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.3310
                                                  Adj R-squared   =     0.2560
                                                  Within R-sq.    =     0.0443
Number of clusters (gvkey)   =      2,521         Root MSE        =     0.4546

                              (Std. Err. adjusted for 2,521 clusters in gvkey)
------------------------------------------------------------------------------
             |               Robust
      capex1 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         big |  -.0903941   .0550729    -1.64   0.101    -.1983867    .0175986
        lage |  -.4259193   .0227298   -18.74   0.000    -.4704902   -.3813484
    lage_big |   .0546331   .0192488     2.84   0.005     .0168881    .0923782
------------------------------------------------------------------------------
```

Interaction term: example of mis-specification

Run clean triple-diff

$$\frac{I_{i,t}}{PPE_{i,t-1}} = log(Age_{i,t}) \times [Big_{i,t0}] + log(Age_{i,t}) + \alpha_i + \delta_t \times [Big_{i,t0}]$$

Remember that we should have:

- $\beta^{small} = -0.5486915$
  - -0.5486915

$\Rightarrow$ Marginal effect = $= 0.2992922$
  - 0.2992922

```
. reghdfe capex1 big lage lage_big , a(gvkey year) vce(cluster gvkey)
(dropped 38 singleton observations)
(converged in 8 iterations)

HDFE Linear regression                            Number of obs   =     25,172
Absorbing 2 HDFE groups                           F(   3,   2520) =     117.91
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.3310
                                                  Adj R-squared   =     0.2560
                                                  Within R-sq.    =     0.0443
Number of clusters (gvkey)   =      2,521         Root MSE        =     0.4546

                           (Std. Err. adjusted for 2,521 clusters in gvkey)
------------------------------------------------------------------------------
             |               Robust
      capex1 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         big |  -.0903941   .0550729    -1.64   0.101    -.1983867    .0175986
        lage |  -.4259193   .0227298   -18.74   0.000    -.4704902   -.3813484
    lage_big |   .0546331   .0192488     2.84   0.005     .0168881    .0923782
------------------------------------------------------------------------------
```

```
. reghdfe capex1 big lage lage_big , a(gvkeybig yearbig) vce(cluster gvkey)
(dropped 205 singleton observations)
(converged in 10 iterations)
note: big omitted because of collinearity

HDFE Linear regression                            Number of obs   =     25,005
Absorbing 2 HDFE groups                           F(   2,   2516) =     150.84
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.3591
                                                  Adj R-squared   =     0.2714
                                                  Within R-sq.    =     0.0403
Number of clusters (gvkey)   =      2,517         Root MSE        =     0.4458

                           (Std. Err. adjusted for 2,517 clusters in gvkey)
------------------------------------------------------------------------------
             |               Robust
      capex1 |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         big |          0  (omitted)
        lage |  -.5486915   .0367461   -14.93   0.000    -.6207472   -.4766358
    lage_big |   .2992922   .0455816     6.57   0.000     .2099108    .3886736
------------------------------------------------------------------------------
```

Outline

How to identify a financially constrained firm?

Two lectures on that:

Identifying Financially Constrained Firms: Methodology

Identifying Financially Constrained Firms: Data

Thank you!

Adrien Matray adrien.matray@gmail.com